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RESEARCH ARTICLE

Control of Movement

The effects of electrode configuration and location on motor thresholds, tolerance, and muscle selectivity of transcutaneous spinal cord stimulation in stroke

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Abstract

Identifying optimal conditions for transcutaneous spinal cord stimulation (tSCS), such as electrode configuration, is essential for clinical translation, particularly as this promising neuromodulation therapy expands to include diverse neurological populations. However, before clinical application, it is important to understand how different configurations affect dosage and muscle recruitment. We define dosage as the percentage of the current needed to elicit a motor response (the resting motor threshold, or RMT) of the current value tolerated during walking when stimulation is delivered continuously. Another consideration is whether electrode configurations can selectively recruit specific muscles, potentially guiding tSCS protocols to prioritize the muscles most critical for functional improvement. This study tested five electrode configurations in individuals with chronic stroke: one round electrode at T11-T12 vertebrae, one round electrode at L1-L2, two separate round electrodes at T11-T12 and L1-L2 from separate current channels, two separate round electrodes at T11-T12 and L1-L2 from the same current channel, and a rectangular electrode spanning T11-T12 and L1-L2. No single electrode configuration clearly outperformed others in terms of tolerated dosage, with most supporting stimulation near 50% of RMT except the two-channel setup with separate current channels, which showed lower dosage (∼35% RMT). Additionally, muscle selectivity did not significantly differ across configurations. These findings inform stimulation dosage feasibility for future tSCS protocols using different electrode configurations and the need for future studies to evaluate whether current dosage levels are adequate to induce meaningful neuromodulation and clinical benefits in the stroke population.

NEW & NOTEWORTHY This is the first study to examine how electrode configuration influences stimulation comfort in transcutaneous spinal cord stimulation (tSCS) relative to resting motor threshold (RMT) in individuals with stroke. Across five configurations, most were tolerated at ∼50% RMT, whereas a two-channel setup reduced tolerance (∼35%). Muscle selectivity did not differ significantly among configurations. These findings establish critical feasibility benchmarks for electrode placement in future tSCS protocols for stroke rehabilitation.

neuromodulation; parameter selection; rehabilitation; spinal cord stimulation; spinal reflexes

INTRODUCTION

Transcutaneous spinal cord stimulation (tSCS) is an increasingly popular noninvasive neuromodulation therapy being applied for motor rehabilitation in individuals with neurological impairments, including stroke (1–3). However, optimizing tSCS remains challenging because of the many stimulation parameters that can be manipulated, such as stimulation waveform, pulse frequency, carrier frequency, electrode size, location, and stimulation intensity (4). A critical step in translating tSCS to clinical applications is understanding how stimulation parameters affect muscle recruitment and

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patient comfort, thereby informing stimulation protocols to maximize clinical feasibility and efficacy.

One factor that may influence the spatial distribution of current and subsequent neural activation resulting from tSCS is electrode configuration [i.e., the number, geometry, and site(s) of electrodes along the spine]. Prior studies in healthy individuals suggest that varying electrode configuration may result in different resting motor thresholds (RMTs) for the same muscle (5). Additionally, varying placement along the rostro-caudal axis may result in differential recruitment of proximal and distal lower limb muscles according to the anatomical organization of motoneuron pools innervating leg muscles (6–8), and utilizing multielectrode configurations may enhance muscle selectivity compared to a single electrode (9). However, these effects have not yet been investigated systematically in individuals post stroke. Given altered motoneuron excitability in these individuals (10), it is unclear whether these recruitment patterns observed in healthy individuals are preserved in people with stroke.

In addition to evaluating muscle selectivity elicited by tSCS in individuals with stroke, it is essential to understand how electrode configuration influences stimulation tolerance, as this affects the feasible stimulation dosage that can be used during walking-based interventions. Stimulation tolerance refers to the highest intensity a participant can comfortably tolerate without pain and serves as a practical upper limit for safe stimulation. In neurological populations where sensation is preserved, discomfort during stimulation may limit the maximum tolerable current intensity, potentially arising from both sensory activation at the stimulation site and paraspinal muscle contraction (11). To understand how comfort impacts dosing, stimulation tolerance can be measured relative to each individual's resting motor threshold (RMT). The RMT is the minimum current intensity required to evoke a muscle contraction, and it is influenced by individual physiological factors (e.g., body composition, neurological impairment) and stimulation parameters (11–15). The percentage of RMT tolerated by participants during continuous stimulation may serve as a measure of dosage; however, how different electrode configurations influence this relationship in chronic stroke remains unknown. If certain configurations increase the percentage of RMT that can be tolerated, they may enable delivery of higher intensities that could improve therapeutic efficacy. In stroke rehabilitation, we hypothesize that stimulation should be subthreshold to promote voluntary motor control without triggering involuntary muscle contractions; however, the optimal subthreshold level remains unclear, making it important to understand not only whether a dose is subthreshold but how far below threshold it should be to provide a therapeutic benefit. Identifying configurations that allow higher tolerance relative to RMT could therefore provide a framework to explore subthreshold dosing options, guide the selection of stimulation parameters that maximize dosage, and help determine an optimal intensity for long-term tSCS interventions.

This study systematically evaluated the effects of electrode configuration on spinal motor evoked responses (sMERs) and stimulation tolerance in individuals post stroke. Specifically, we examined the effects of electrode location along the lumbar enlargement (T11, L1, T11 + L1 vertebrae), number of electrodes (single vs. multielectrode), geometry of electrode (round vs. rectangular), and current delivery method within multielectrode configurations (single channel splitter vs. 2 separate channels). By understanding how different configurations influence RMT, comfort, and muscle recruitment selectivity, this study provides insights into optimizing tSCS protocols for lower extremity rehabilitation. Additionally, we explored whether findings from healthy populations regarding recruitment selectivity, i.e., the ability of different electrode configurations to preferentially activate certain muscles over others, could be replicated in individuals post stroke, further informing stimulation strategies for this clinical population. Overall, the purpose of this study was to understand how different configurations vary in terms of comfortable delivery of near-threshold stimulation, enabling adequate parameter selection for subthreshold application in future clinical settings.

METHODS

Participants

Seventeen participants with stroke were recruited by convenience sampling from the Shirley Ryan AbilityLab (Table 1). Participant inclusion criteria included the following: aged over 18 yr, at least 6 mo post stroke, hemiplegia secondary to a single stroke, Functional Ambulation Category of 2 or greater, able to provide informed consent, not currently receiving physical therapy services, and physician approval. Participant exclusion criteria were the following: ataxia, history of multiple strokes, botulinum toxin injection in the lower extremity within the last 4 mo, Modified Ashworth Scale of 3 or greater in the lower extremity, pregnancy or nursing, presence of pacemaker, active pressure sores, unhealed bone fractures, peripheral neuropathies, painful musculoskeletal dysfunction due to active injuries or infections, severe contractures, medical illness limiting ability to walk, active urinary tract infection, clinically significant depression, psychiatric disorders or ongoing drug abuse, metal implants in spine, and active cancer or cancer remission $<$ 5 yr.

Table 1. *Participant demographics*

ID	Sex	Age, yr	Chronicity, yr	Stroke Type	Height, cm	Weight, kg	SSV, m/s
1	F	59	5	Isch	163	82.7	0.77
2	M	56	9	Isch	175	118.2	0.96
3	M	70	9	Hemo	173	80.9	0.27
4	M	55	5	Isch	173	79.5	0.46
5	M	57	8	Isch	168	74.1	0.68
6	F	73	3	Isch	165	63.5	0.81
7	M	71	6	Hemo	175	61.4	0.09
8	M	63	7	Isch	190	123.2	0.79
9	M	52	6	Hemo	182	115.9	1.10
10	M	49	1	Hemo	175	104.5	1.56
11	F	64	1	Hemo	160	84.1	0.37
12	M	41	2	Hemo	173	79.4	0.80
*13**	F	46	5	Hemo	164	92.5	0.50
*14**	M	65	5	Isch	170	82.7	0.29
*15**	M	54	2	Isch	175	80.5	1.08
*16**	M	48	7	Hemo	172	76.4	0.67
*17**	F	54	1	Isch	170	88.6	1.18

*The last 5 participants listed did not achieve a resting motor threshold (RMT) of at least 1 muscle for every electrode configuration during the spinal motor evoked responses tests. Hemo, hemorrhagic; Isch, ischemic; SSV, self-selected velocity.

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This study was conducted at Shirley Ryan AbilityLab in Chicago, Illinois. The trial protocol was approved by the Northwestern University Institutional Review Board (STU00215009). All participants provided written informed consent. The study design and conduct complied with all relevant regulations regarding the use of human study participants, and the study was conducted in accordance with the criteria set by the Declaration of Helsinki. The trial protocol was preregistered at ClinicalTrials.gov (NCT05167786).

Study Protocol

Five different configurations of electrode site and size were tested (see Fig. 1): *1*) one round 3.2-cm-diameter electrode (LotFancy; Wuxi Jiajian Medical Instruments Co., Ltd., China) between the T11 and T12 spinous processes (referred to as T11), *2*) one round 3.2-cm-diameter electrode between the L1 and L2 spinous processes (referred to as L1), *3*) two separate round electrodes between the T11 and T12 and L1 and L2 spinous processes, respectively, connected to two separate channels (referred to as 2CH), *4*) two separate round electrodes between the T11-T12 and L1-L2 spinous processes, respectively, connected to a single channel with a splitter cable (referred to as SPLIT), and *5*) one 5 × 10-cm rectangular electrode (AcuZone, Los Angeles, CA) with the long edge spanning from T11 vertebra to L1 (referred to as RECT). 2CH and SPLIT differed in the way the electrodes were connected to the current source. The splitter setup involved a single cable from the current source that split into two separate electrodes for each location. For 2CH, each site received stimulation from two separate channels delivered with a 1-ms lag, where T11 was delivered first and then L1. For all configurations, the stimulating electrodes over the spinal cord were positioned along the midline, and two return electrodes (5 × 10 cm; AcuZone, Fallbrook, CA) were placed symmetrically over the anterolateral aspect of the iliac crests.

Spinal Motor Evoked Responses

Spinal motor evoked responses (sMERs) are evoked compound muscle action potentials recorded with surface electromyography (EMG) that result from stimulating the spinal cord with single pulses. For each configuration, the electrode(s) were placed at the sites described in Fig. 1. As the participants lay supine and relaxed, a constant-current stimulator (DS8R; Digitimer Ltd., United Kingdom) was used to deliver stimulation as a 2-ms square, symmetric biphasic pulse (1 ms per phase, cathodic followed by anodic pulse with a 1-μs interphase interval) for 5 trials per intensity at an interval of 5 s apart (Fig. 2*A*). Stimulation was delivered as single pulses in 10-mA intensity increments up to a maximum of 250 mA, or until the participant verbally expressed discomfort (this is referred to as single-pulse tolerance). sMERs were recorded with surface EMG electrodes (bipolar Ag-AgCl; GS26, Bio-Medical Instruments, Clinton Charter Township, MI) placed longitudinally on the muscle bellies of rectus femoris (RF), medial hamstring (HAM), tibialis anterior (TA), and medial gastrocnemius (MG) on both legs. Ground reference electrodes for the EMG were placed bilaterally over the bony prominence of the patella. EMG signals were sampled at 4,000 Hz with a PowerLab 16/35 data acquisition system operated with LabChart software (v7.2; ADInstruments, Bella Vista, NSW, Australia). Afterward, the EMG data were low-pass filtered (5th-order low-pass Butterworth, 450 Hz) with MATLAB (MathWorks Inc, United States).

From this assessment, the resting motor threshold was calculated for each muscle for each configuration. The RMT was defined as the minimum current intensity (measured in mA) required to obtain a muscle response with a peak-to-peak amplitude of at least 0.05 mV in three out of the five pulses (Fig. 2*B*). Additionally, the single-pulse tolerance (measured in mA) was recorded in this assessment as the maximum current the participant could tolerate when doing sMERs.

Continuous Stimulation Tolerance Testing

In each configuration, the tolerance to continuous stimulation was tested as participants walked on the treadmill to emulate the clinical intervention setting (Fig. 2*C*). The participants walked on the treadmill at their comfortable speed as continuous stimulation (2-ms square, symmetric biphasic pulse delivered at 30 Hz) was slowly increased until the participant reported discomfort. Then, to test habituation to stimulation, the stimulation intensity was decreased by 10% and the participant walked with this intensity for 2 min. The stimulation was once again slowly increased until participants reported that the stimulation was too intense. This final current intensity value was recorded as the continuous tolerance intensity (measured in mA).

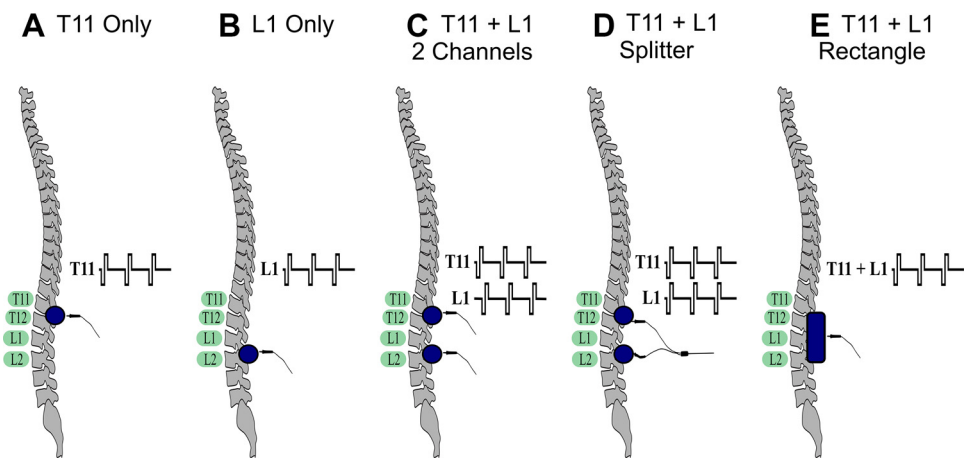


Figure 1. Schematic of electrode configurations. *A*: T11 only: single-site 3.2-cm round electrode between the T11 and T12 spinous processes. *B*: L1 only: single-site 3.2-cm round electrode between the L1 and L2 spinous process. *C*: 2 channels (2CH): two 3.2-cm round electrodes on separate channels between the T11–T12 and L1–L2 spinous processes, delivered 1 ms apart. *D*: splitter (SPLIT): two 3.2-cm round electrodes on single channel via splitter cable between the T11–T12 and L1–L2 spinous processes. *E*: rectangle (RECT): 5 × 10-cm rectangular electrode spanning T11 to L2 spinous processes.

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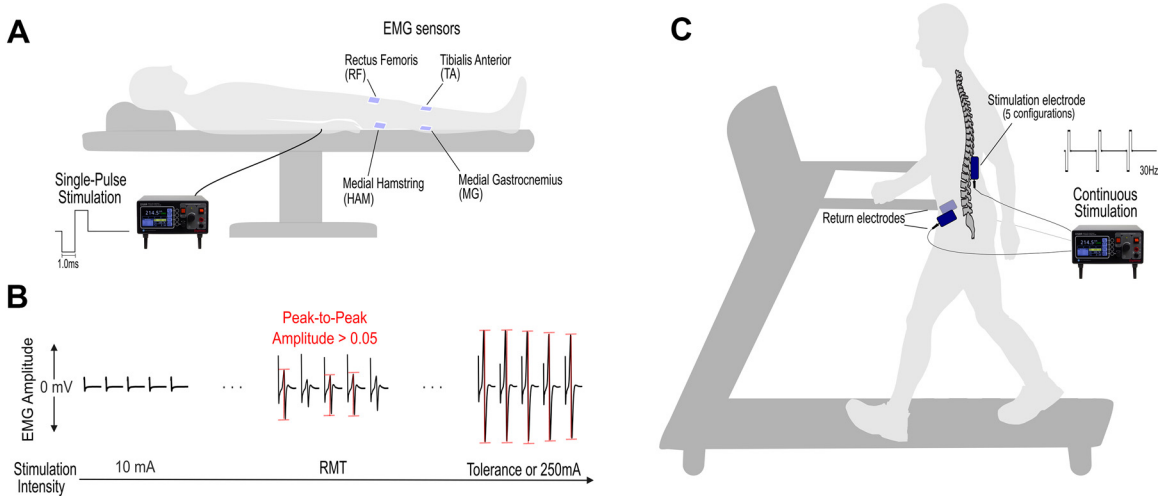


Figure 2. Study protocol. *A*: protocol for spinal motor evoked responses (sMERs). Single pulses are delivered at incremental intensities while electromyography (EMG) sensors record muscle responses bilaterally at rectus femoris (RF), medial hamstring (HAM), tibialis anterior (TA), and medial gastrocnemius (MG) with the purpose of obtaining the resting motor threshold (RMT). *B*: example of muscle responses recorded by EMG as stimulation intensity increases in the sMER protocol. RMT is defined as the lowest stimulation that elicits a peak-to-peak muscle response of at least 0.05 mV. *C*: the maximum tolerance to stimulation is obtained as participant walks on the treadmill and stimulation is delivered at 30 Hz with each configuration.

Each configuration was used for both the sMER assessment and continuous tolerance testing. For each participant, all configurations were tested within a single session in randomized order. The session began with three sMER tests followed by three tolerance tests. The remaining two configurations were then tested with sMERs, concluding with two final tolerance tests.

Data Analysis, Outcomes, and Statistics

Inclusion.

Out of seventeen participants enrolled, five participants did not exhibit a measurable muscle response (e.g., an RMT) in any muscle for one or more of the five configurations tested (see Supplemental Table S1 for details). Therefore, only participants who obtained at least one RMT in all five configurations ($n = 12$) were included in the analysis of RMT, selectivity, and percentage of RMT tolerated. As the tolerance analysis did not depend on RMTs and was completed by all participants, all $n = 17$ participants were included in the analysis of single-pulse tolerance and continuous tolerance.

Resting motor thresholds.

The resting motor thresholds (RMTs) were compared across the five electrode configurations to assess the influence of site and electrode size on absolute threshold differences. Each participant had a maximum of eight RMTs, corresponding to each muscle measured, for each configuration. A paired t test was performed between paretic (P) and nonparetic (NP) RMTs, and it was determined that a single average across the eight muscles was appropriate (see Supplemental Table S2). Because of the nonparametric nature of the data, a Friedman test was employed to identify significant differences in RMTs across configurations. This test accounts for the repeated-measures design by considering the within-subject variability. If there was significance, post hoc comparisons were conducted to determine pairwise differences between configurations. Statistical significance was set at $P < 0.05$, and Bonferroni corrections were used.

Tolerance to stimulation.

Continuous and single-pulse tolerance to stimulation were compared across the five electrode configurations with the same analytical approach as for the RMTs. For each participant, tolerance values were compared across configuration with a Friedman test. Additionally, a Friedman test evaluated the effect of stimulation order on tolerance, testing whether the highest tolerance typically occurred with the final configuration. Supplemental Table S3 details the testing order of stimulation configurations for each participant.

Pearson correlation analyses were performed to assess the relationships between RMT and stimulation tolerance relative to participant characteristics (age, chronicity, weight, height, and walking speed), consistent with Veit et al. (12). One participant (*ID 8*) met a predefined outlier criterion (values exceeding 1.5 the interquartile range) and was excluded from the correlation analysis. These results are reported in Supplemental Table S4.

Percentage of RMT tolerated.

The percentage of RMT tolerated provides a person- and configuration-specific normalized measure of how comfortable and effective each configuration is relative to the current needed to elicit a response (RMT). The tolerance to stimulation was normalized to the average RMT with the following formula:

$$100 \times \frac{\text{Continuous Tolerance}}{\text{Average RMT}}$$

The same statistical approach described for comparing RMT and tolerance across configurations was employed here.

Muscle selectivity.

Muscle selectivity is described as preferential recruitment of specific muscles over others depending on electrode configuration. To assess muscle selectivity, the RMTs of each of the eight muscles recorded (4 on each side: HAM,

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RF, TA, and MG) were ranked. RMTs represent the minimum current needed to elicit a response; thus, ranking these values within each configuration can elucidate muscle selectivity. For each participant and configuration, muscles were assigned a rank from 1 to 8, with 1 indicating the first muscle activated (lowest RMT) and 8 representing the last muscle activated (highest RMT) within that particular configuration. If two or more muscles had the same RMT, they were assigned the average rank (e.g., if ranks 7 and 8, then 7.5). Muscles that did not exhibit an RMT for a particular electrode configuration were assigned the highest rank. Rankings from the left and right sides were then averaged to obtain a ranking for each general muscle: HAM, MG, TA, and MG.

This ranking enabled us to compare the order of muscle activation within each configuration and to determine whether different configurations influenced muscle recruitment order, potentially indicating selectivity. Two analyses were performed: first, Friedman tests were conducted within each configuration to determine whether the order of muscle activation differed significantly. Second, Kendall's tau correlation coefficient was calculated for each participant to assess the similarity in activation order across configurations, resulting in nine pairwise comparisons. Kendall's tau is a nonparametric statistic (−1 to 1) to quantify agreement in activation order between configurations. A coefficient closer to 1 indicates strong similarity in activation order, whereas a coefficient near 0 suggests little to no consistent relationship. A coefficient near −1 would indicate a largely reversed activation order between configurations. Corresponding Kendall's tau correlation coefficients and their P values were obtained, and both metrics were averaged across participants to assess general trends. Statistical significance was set at $P < 0.05$.

Muscle specificity.

Simultaneous recruitment of multiple muscles at the same intensity reduces muscle specificity and can influence selectivity rankings by producing ties in recruitment order. To explicitly quantify muscle specificity (defined as the number of muscles recruited at the same intensity), for each participant and stimulation configuration the minimum RMT value was identified across all recorded muscles and the number of muscles recruited at that value was quantified. Then, for each configuration, we summarized the proportion of participants exhibiting activation of a single muscle versus multiple muscles at their lowest RMT.

RESULTS

Resting Motor Thresholds

The resting motor thresholds (RMTs) were compared across the five electrode configurations (Fig. 3*A*). A Friedman test revealed a significant difference in RMTs across configurations [$Q(4) = 17.6$, $P < 0.001$, Kendall's $W = 0.37$, $N = 12$]. Median raw RMT values were as follows (in ascending order): 2CH: 71 mA [interquartile range (IQR): 51–81], RECT: 71 mA (IQR: 58–98), L1: 82 mA (IQR: 60–90), T11: 88 mA (IQR: 72–102), and SPLIT: 89 mA (IQR: 68–105). The 2CH configuration had the lowest thresholds out of all configurations and was significantly lower than T11 ($P = 0.005$). As previously reported by Veit et al. (12), RMTs were significantly correlated with participant weight but not height, age, or chronicity (see Supplemental Table S4).

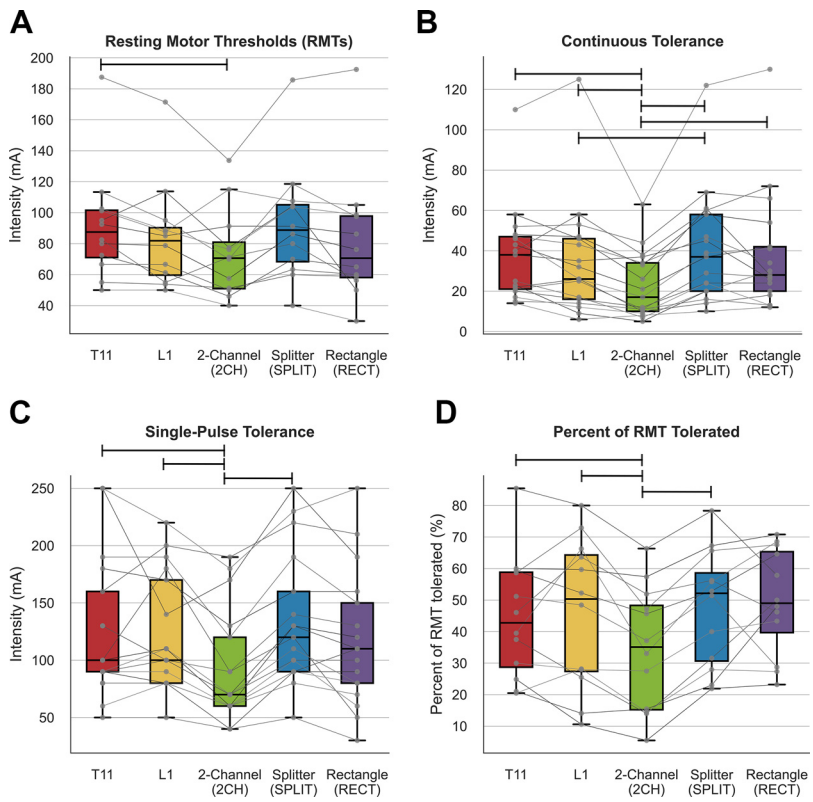


Figure 3. *A*: resting motor threshold (RMT) differences between configurations ($N = 12$ participants). *B*: maximum intensity tolerated as stimulation was delivered continuously at 30 Hz while participants walked on the treadmill ($N = 17$ participants). *C*: single-pulse tolerance obtained as the maximum stimulation tolerated during the spinal motor evoked response (sMER) assessment ($N = 17$ participants). *D*: tolerance was normalized to RMT to understand the percentage of RMT that could be delivered per configuration ($N = 12$ participants). Bars represent statistical significance ($P < 0.05$).

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Tolerance

All 17 participants were considered for the continuous tolerance to stimulation analysis. Tolerance was obtained as participants walked on the treadmill and the highest value they could tolerate after 2 min of walking was recorded. Median continuous tolerance (Fig. 3*B*) was higher for T11 (median: 38 mA, IQR: 21–47), followed by SPLIT (median: 37 mA, IQR: 20–58), RECT (median: 28 mA, IQR: 20–42), and L1 (median: 26 mA, IQR: 16–46), and least tolerable was 2CH (median: 17 mA, IQR: 10–34). 2CH was significantly lower compared to all other four configurations: SPLIT ($P < 0.001$), T11 ($P < 0.001$), L1 ($P < 0.001$), and RECT ($P = 0.015$). Additionally, SPLIT was significantly higher than L1 ($P = 0.032$). Single-pulse tolerance (Fig. 3*C*) was lowest for 2CH (median: 70 mA, IQR: 60–120) and significantly different from T11 ($P = 0.017$), L1 ($P = 0.005$), and SPLIT ($P < 0.001$). Tolerance to continuous stimulation was not correlated with any participant characteristics (see Supplemental Table S4).

Additionally, the effects of order were tested to explore whether continuous tolerance generally increased over the course of the session. The Friedman's test did not find a difference in tolerance corresponding to the order of configuration tested [$Q(4) = 2.06$, $P = 0.725$, Kendall's $W = 0.03$, $N = 17$]. However, within each configuration, a habituation period of 2 min allowed tolerance to increase ~25 ± 9% on average across all configurations, highlighting that tolerance within the same configuration can increase as participants acclimate to the stimulation.

Percentage of RMT Tolerated

The percentage of RMT tolerated (Fig. 3*D*) was lowest for 2CH (median: 35%, IQR: 15–48) and significantly different from SPLIT (median: 52%, IQR: 31–59, $P = 0.005$), L1 (median: 50%, IQR: 27–64, $P = 0.015$), and T11 (median: 43%, IQR: 29–59, $P = 0.034$). RECT median percentage was 49% (IQR: 40–65).

Muscle Selectivity

Friedman tests revealed no significant difference between the order of muscles recruited within configuration (T11 $P = 0.523$; L1 $P = 0.290$; 2CH $P = 0.871$; SPLIT $P = 0.271$; RECT $P = 0.699$) (Fig. 4).

Ranking agreement was assessed with Kendall's tau across the five configurations for each participant. The strongest agreement, and only pairs showing significance for at least of 50% of the participants, was found between T11 and SPLIT [$\tau = 0.702$ (IQR: 0.504–0.802), $P = 0.034$ (IQR: 0.011–0.127)] and between L1 and RECT [$\tau = 0.715$ (IQR: 0.384–0.810), $P = 0.037$ (IQR: 0.015–0.251)].

Muscle Specificity

Configurations with a higher proportion of participants recruiting only one muscle at threshold were interpreted as demonstrating greater muscle specificity. With this metric, 2CH, RECT, and T11 showed the highest specificity, with at least 50% of participants recruiting a single muscle at their lowest RMT (Fig. 5). In contrast, SPLIT and L1 more frequently elicited concurrent activation of multiple muscles at threshold, indicating reduced specificity.

To further characterize recruitment behavior, we tabulated the frequency of the first-recruited muscle across participants for each configuration, restricting the analysis to subjects who exhibited activation of only a single muscle at the lowest RMT (Supplemental Table S5). For example, in the RECT configuration, eight participants showed that they recruited one muscle at lowest RMT. For those eight, initial recruitment most commonly occurred in the tibialis anterior (TA) and medial gastrocnemius (MG) for a total of five participants.

Importantly, high specificity at threshold did not necessarily correspond to consistent selectivity across participants. For instance, although the RECT configuration often activated only a single muscle at the lowest RMT (indicating

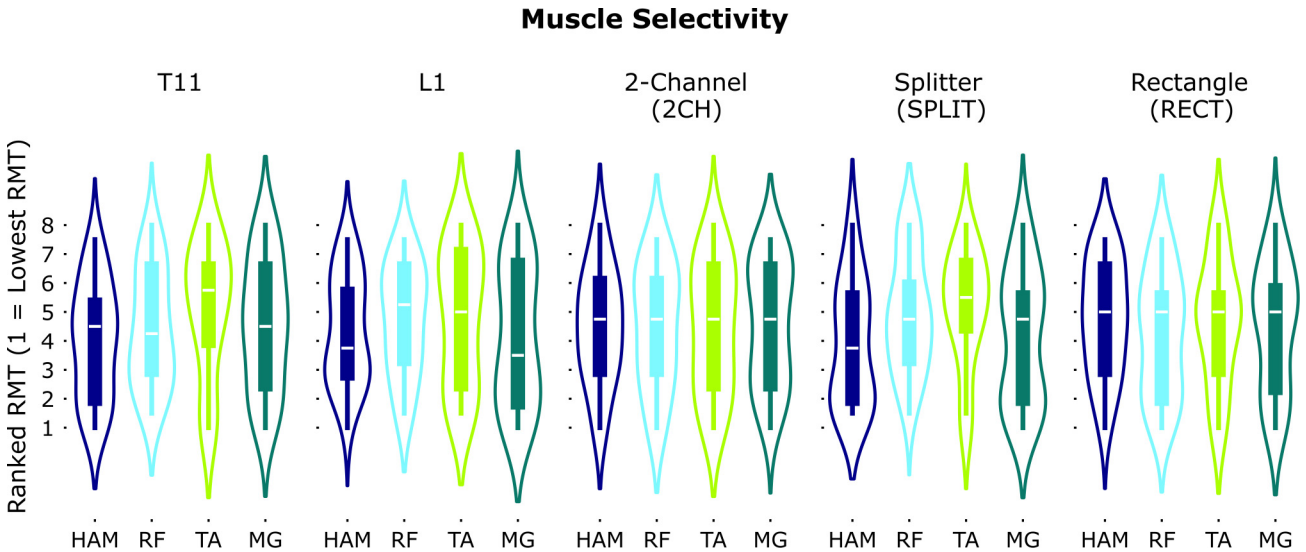


Figure 4. Rankings of the order in which muscles were activated per configuration ($N = 12$ participants). For each participant and configuration, the resting motor thresholds (RMTs) for each of the 8 muscles were ranked in ascending order from 1 to 8, where 1 represents the muscle with the lowest RMT and therefore the muscle recruited first. The right and left muscle rankings were averaged. HAM, medial hamstring; MG, medial gastrocnemius; RF, rectus femoris; TA, tibialis anterior.

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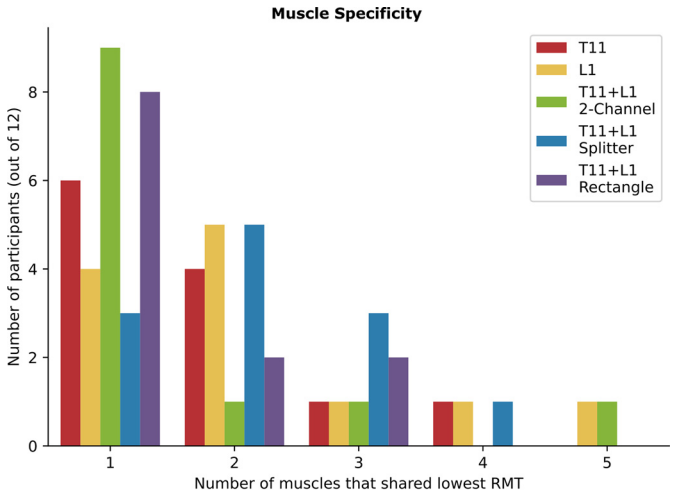


Figure 5. Muscle recruitment specificity across electrode configuration at lowest resting motor threshold (RMT). The *x*-axis represents the number of muscles that were activated at the lowest RMT, and the *y*-axis indicates the number of participants (out of 12), grouped by configuration. Greater specificity is reflected by configurations that preferentially activate a single muscle rather than multiple muscles at the same intensity.

relatively high specificity), the identity of that first-recruited muscle varied considerably between participants, reflecting lower interparticipant consistency in recruitment order (Fig. 4). Conversely, L1 and SPLIT exhibited lower specificity due to more frequent multimuscle coactivation at threshold.

DISCUSSION

This study investigated the way in which electrode configuration influences spinal motor evoked potentials and stimulation tolerance in individuals post stroke. Our goal was to explore how different electrode configurations may affect stimulation dose. Since the optimal dosing of tSCS is unknown, one approach to determining the stimulation intensity to be used during training is using the RMT intensity (obtained through sMERs) as a reference point and quantifying stimulation delivered relative to RMT. However, a key challenge arises when the stimulation intensity tolerated is far below their RMT. This narrows the window of possible stimulation intensities and limits the ability to deliver clinically meaningful doses of stimulation. In individuals with some walking function, such as individuals post stroke, it is suggested to deliver stimulation just below threshold to support descending drive without overriding voluntary movement (16). This requires a careful balance. If a person's tolerance is much lower than their RMT, it becomes difficult to deliver a dose that is both comfortable and effective. Therefore, minimizing the gap between RMT and maximum tolerable intensity is critical since it increases the range of stimulation dosages for testing and options for tailoring tSCS as a therapy. This consideration is especially relevant as tSCS gains popularity beyond the spinal cord injury (SCI) population, where differing rehabilitation goals may require tailored stimulation dosages and preserved sensation can limit the amount of current that can be comfortably delivered.

Our previous research found that, although parameters like carrier frequency and waveform affect both RMT and absolute tolerance level, the percentage of RMT tolerated remains consistent. This indicates that these parameters do not significantly alter perceived comfort when considered relative to the motor threshold (11, 14). For this study, we wanted to understand the same relationship of comfort when varying electrode quantity and location (T11 vs. L1 vs. T11 + L1), geometry (round vs. rectangular), and current delivery method for multisite configurations (2 channels with a single current source with a splitter or 2 separate channels). We determined whether specific electrode configurations could minimize the gap between RMT and tolerance intensity, allowing greater dosing flexibility over stimulation intensity while maintaining patient comfort. Additionally, we explored how electrode site placement influences the selectivity of muscle recruitment in individuals post stroke, providing insight into how tSCS might differentially engage neural pathways in this population.

Among all configurations, 2CH (2 channels) was the most uncomfortable and differed the most across all the metrics measured, including lowest values of RMT and tolerance (both in single pulse and continuous). This is not surprising, as the electrodes are connected to two different current sources where two concurrent pulses are delivered at the same intensity 1 ms apart, as opposed to the SPLIT configuration. The SPLIT configuration effectively distributes the applied current across two electrode sites arranged in parallel, such that the stimulator adjusts voltage to maintain the prescribed total current while current divides between sites according to their relative impedance. As a result, the current displayed by the stimulator does not necessarily reflect the current delivered to each individual electrode, and local electric field strength at each site is reduced compared to single-site and 2CH configurations. This redistribution likely contributed to the higher stimulation thresholds observed for SPLIT. However, thresholds were not doubled, as might be expected if equivalent field strength were required at each site independently. This may suggest a spatial summation effect whereby concurrent subthreshold fields across spinal segments combine to reach neural activation. Additionally, variability in electrode-skin impedance, influenced by factors such as skin hydration and electrode contact quality, may further bias current distribution between sites. Consistent with this, the recruitment pattern for SPLIT more closely resembled that of T11, potentially indicating preferential current flow or greater effective field strength at that segment. Together, these findings highlight the way that current distribution and summative neural recruitment influence both threshold and selectivity in multisite tSCS configurations.

Therefore, two arrangements with visually similar setup, 2CH and SPLIT, that only differ in the current delivery method were significantly different in the percentage of RMT that could be tolerated. Our results demonstrated that the 2CH configuration had the lowest percentage of RMT tolerated, suggesting that this arrangement might limit the amount of current that could be delivered in an intervention to ~35% relative to average RMT. All other configurations tested allowed delivery of ~50% of RMT, which agrees with previous research reporting ~56% in able-bodied individuals with two 3.2-cm-diameter

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electrodes placed paraspinally (although 2CH vs. splitter setup was not specified) (11). This is an important consideration in quantifying dosage, as 2CH may be limited in the amount of current close to the RMT that can be delivered, assuming that maximal subthreshold percentages of RMT might be more effective as a previous study has claimed (17). Future studies are needed to examine the relationship between percentage of RMT delivered and therapeutic efficacy in neurological populations. Additionally, studies using multisite stimulation should clearly report the current delivery configuration, given its impact on tolerability and dosing.

Between the two single-site conditions, T11 had slightly higher absolute tolerance than L1, though both showed similar ranges when normalized to RMT (T11 IQR: 30–60%; L1 IQR: 30–65%). T11 tended to recruit more upper leg muscle activation, whereas L1 showed more lower leg muscle recruitment, aligning with expectations based on spinal segmental anatomy and previous studies (18). However, these trends were not statistically significant and were inconsistent across participants, likely reflecting the inherent variability in the neurological effects of stroke and human anatomy variability (19).

When comparing multisite versus single-site, the RECT configuration demonstrated lower RMTs compared to individual stimulation sites. This observation is consistent with findings reported by Skiadopoulos et al. (5), who similarly observed reduced thresholds with rectangular electrodes relative to single-site stimulation. Additionally, the RECT configuration, which covers both T11 and L1, exhibited recruitment patterns more similar to L1. This may reflect a more diffuse electrical field distribution across the larger contiguous electrode surface, analogous to the continuous hydrogel configuration described by Krenn et al. (6) or the rectangular configuration used by Bryson et al. (9), which reduced muscle selectivity by permitting longitudinal current spread. Therefore, the current distribution of the RECT may preferentially activate spinal regions with lower activation thresholds, consistent with the generally lower thresholds at L1 compared to T11. In contrast, T11 and SPLIT configurations showed strong agreement in muscle recruitment order from Kendall's tau analysis, suggesting that in the SPLIT setup where there is a single channel splitting between T11 and L1 sites, more current may be directed toward T11, therefore mimicking T11 recruitment as discussed above. However, these assumptions require further investigation through modeling studies, as current flow likely varies between individuals and the observed differences in selectivity and muscle recruitment patterns were not statistically significant. One limitation of this study was the absence of a control group, making it difficult to determine whether the observed effects were due solely to electrode configuration or to a combination of electrode configuration and underlying neurological conditions. A previous study comparing electrode configurations for upper extremity stimulation at the cervical level in individuals with SCI found that recruitment patterns in SCI did not match those of control individuals (20). The differences were attributed to central nervous system adaptation following the injury. Similarly, our findings did not show clear muscle selectivity compared to studies involving neurologically intact participants (7, 8), suggesting that the underlying neurological condition may have influenced recruitment patterns. Additionally, electrode placement relied on surface palpation of anatomical landmarks, and, although performed by trained physical therapists, precise targeting across participants is not perfectly objective and may have contributed to the observed variability in selectivity.

Despite these limitations, our results highlight how different electrode configurations may interact with altered neurophysiology and how seemingly similar setups, such as SPLIT and 2CH, can differ significantly in terms of comfort and motor thresholds. From a research perspective, understanding the relationship between RMT and tolerance is critical for the future development of effective tSCS protocols, especially in populations where sensation is preserved and when stimulation is delivered subthreshold to facilitate voluntary motor output. Future studies should explore how percentage of RMT delivered relates to changes in functional outcomes and whether tailoring electrode configurations to rehabilitation goals can enhance therapeutic efficacy.

ETHICAL APPROVALS

This study was conducted in accordance with the principles outlined in the Declaration of Helsinki. Ethical approval was obtained from the Institutional Review Board (IRB) at Northwestern University, ensuring that all procedures adhered to ethical standards for research involving human participants. Written informed consent was obtained from all participants before their involvement in the study. The trial protocol was preregistered at ClinicalTrials.gov (NCT05167786).

DATA AVAILABILITY

The data supporting the findings of this study are openly available in Figshare at https://doi.org/10.6084/m9.figshare.31792210.

SUPPLEMENTAL MATERIAL

Supplemental Tables S1–S5: https://doi.org/10.6084/m9.figshare.31792249.

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DISCLAIMERS

The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

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AUTHOR CONTRIBUTIONS

N.C.V., S.A., A.K., K.A.M., C.Y., and A.J. conceived and designed research; N.C.V., S.A., A.K., K.A.M., and C.Y. performed experiments; N.C.V., S.A., A.K., and M.T. analyzed data; N.C.V., S.A., A.K., and M.T. interpreted results of experiments; N.C.V. and S.A. prepared figures; N.C.V. drafted manuscript; N.C.V., S.A., A.K., K.A.M., C.Y., M.T., E.J.R., R.L.L., and A.J. edited and revised manuscript; N.C.V., S.A., A.K., K.A.M., C.Y., M.T., E.J.R., R.L.L., and A.J. approved final version of manuscript.

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